SAMANTHA ALLEN

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Research Interests

Low-dimensional topology, knot theory in dimensions three and four. In particular, the interplay between classical knot invariants and those arising from Heegaard Floer, and what it can tell us about classical knot-theoretic problems, the knot concordance group, and 3-manifolds.

Academic Appointments

Assistant Professor August 2022 – present
Department of Mathematics and Computer Science
Duquesne University

Limited Term Assistant Professor August 2021 – July 2022
Department of Mathematics
University of Georgia

John Wesley Young Research Instructor July 2018 – July 2021
Department of Mathematics
Dartmouth College

Education

PhD in Mathematics – Indiana University, Bloomington, IN June 2018
Thesis: Relationships between the nonorientable genus and the normal Euler number of nonorientable surfaces whose boundary is a knot
Advisor: Charles Livingston

BS in Mathematics – High Point University, High Point, NC May 2012
BS in Computer Science – High Point University, High Point, NC May 2012

Publications (preprints on following page)

8. S. Allen, C. Livingston, M. Temkin and C.-M. M. Wong, Using knot Floer invariants to detect prime knots. Pacific J. Math. **345** (2026), no. 2, 225-237.
7. S. Allen and C. Livingston, An Upsilon torsion function for knot Floer homology. Math. Res. Lett. **32** (2025), no. 3, 679–705.
6. S. Allen, K. İnce, S. Kim, B.M. Ruppik and H. Turner, Unknotting via null-homologous twists and multi-twists. Pacific J. Math. **330** (2024), no. 1, 25–41.
5. S. Allen, Nonorientable surfaces bounded by knots: a geography problem. New York J. Math. **29** (2023), 1038–1059.
4. S. Allen and J.H. Swenberg, Do Link Polynomials Detect Causality In Globally Hyperbolic Spacetimes? J. Math. Phys. **62** (2021), no. 3, 032503.
3. S. Allen and C. Livingston, Unknotting with a single twist. Enseign. Math. **66** (2020), no. 3-4, 541-589.
2. S. Allen, Concordances from differences of torus knots to L–space knots. Proc. Amer. Math. Soc. **148** (2020), no. 4, 1815–1827.
1. S. Allen, Using secondary Upsilon invariants to rule out stable equivalence of knot complexes. Algebr. Geom. Topol. **20** (2020), no. 1, 29–48.

Preprints

2. S. Allen, J. Isaac, C. Moscariello, R. Muth, B.D. Uwase, and L. Walton, Kleshchev multipartitions, affine Mirković-Vilonen polytopes, and representations of KLR algebras in typr $\mathtt{A}_1^{(1)}$. Preprint: https://arxiv.org/abs/2606.00421

1. S. Allen and C. Livingston, Knot primality: knot Floer homology, metacyclic representations and twisted homology. Preprint: arxiv.org/abs/2508.08102

Awards and Grants

Open Education Teaching Award (Gumberg Libaray at Duquesne University)	Spring 2026
NSF LEAPS-MPS Award DMS-2532488	2025–2027
DU Faculty Development Fund (joint with Robert Muth)	2025–2027
AIM SQuaRE	Summers 2023, 2024
Wimmer Family Foundation summer stipend	Summer 2023
Wimmer Family Foundation award	Fall 2022
MSRI (now SLMSI) 2022 Summer Research in Mathematics	Summer 2022
NSF-AWM Travel Grant	Fall 2019
Joseph & Francis Morgan Swain Fellowship, Indiana University	Spring 2018
IUB Provost's Travel Award for Women in Science, Indiana University	Fall 2017
Rothrock Teaching Award (for excellence in teaching), Indiana University	Spring 2015
Muriel Adams Stahl Fellowship, Indiana University	Fall 2012

Selected Invited Talks

Franciscan University of Steubenville Math Seminar	October 2026
Cornell Topology Festival	May 2026
Philadelphia Area (Contact) Topology (PACT) Seminar	January 2026
University or Nevada, Reno Mathematics Colloquium	October 2025
University of Pittsburgh Algebra, Combinatorics, & Geometry Seminar	February 2024
Barbara T. Faires Allegheny Mountain Colloquium (MAA section)	October 2023
Special Session on Progress in low-dimensional topology (AWM Research Symposium)	October 2023
Virginia Commonwealth University Geometry Seminar	October 2022
PATCH Seminar at Temple University	September 2022
Georgia Topology Conference	May 2022
Georgia Tech Topology Seminar	April 2022
AMS Special Session on Knots, Links, 3-manifolds, and 4-manifolds (JMM)	April 2022
UC Davis Geometry & Topology Seminar	May 2021
Rutgers Topology Seminar	April 2021
Michigan State University Geometry & Topology Seminar	March 2021
AMS Special Session on Low Dimensional Topology (JMM)	January 2021
Boston College Geometry/Topology/Dynamics Virtual Seminar	November 2020
University of Nevada, Reno Algebraic and Geometric Topology Seminar	November 2020
Topic group on Knots, surfaces, and 4-manifolds (NCNGT conference)	June 2020
AMS-AWM Special Session on Women in Topology (JMM)	January 2020
Invariants of Knots, Links, and Low-dimensional Manifolds (AMS Sectional)	October 2019
Knot concordance and low-dimensional manifolds, Le Croisic, France	June 2019
Invariants of Knots, Links, and Low-dimensional Manifolds (AMS Sectional)	April 2019
Algebraic, Combinatoric, and Quantum Invariants of Knots and Manifolds (AMS Sectional)	March 2018
University of Virginia Geometry Seminar	February 2018
University of Oregon Topology Seminar	January 2018

Teaching Experience

For each of the following, I was the **instructor of record**.

Instructor, Duquesne University

Math 116 Calculus II	Spring 2024
Math 215 Calculus III	Falls 2022, 2023; Spring 2023
Math 311 Number Theory	Falls 2022, 2023
Math 250 Foundations of Higher Mathematics	Fall 2025
Math 330W Fundamentals of Geometry	Spring 2026
Math 350W History of Mathematics	Springs 2023-2025
Math 411W Abstract Algebra I	Falls 2024, 2025
Math 412W Abstract Algebra II	Spring 2026
Math 423W Topology	Spring 2025

Instructor, University of Georgia

Math 2250 Calculus I for Science and Engineering	Fall 2021
Math 2260 Calculus II for Science and Engineering	Spring 2022

Instructor, Dartmouth College

Math 8 Calculus of Functions of One and Several Variables	Fall 2018; Spring 2019, 2021
Math 17 Intro to Math Beyond Calculus (*taught remotely*)	Spring 2020
Math 22 Linear Algebra with Applications	Fall 2018
Math 25 Number Theory	Fall 2019
Math 31 Topics in Algebra (*taught remotely in Fall 2020*)	Fall 2019, 2020

Instructor, Indiana University

Math D116 Introduction to Finite Mathematics I	Fall 2017
Math D117 Introduction to Finite Mathematics II	Fall 2015, 2016; Spring 2015
Math J010 Introduction to Algebra	Summer 2013, 2014, 2016
Math J110 Introductory Problem Solving	Summer 2015
Math J111 Introduction to College Math I	Fall 2013, 2014
Math M014 Basic Algebra	Spring 2014
Math T101 Mathematics for Elementary Teachers	Spring 2017

Mentorship

Advisor of undergraduate research projects in knot theory for 3 students — Ongoing

Co-advisor for 3 students in DU's Undergraduate Research Program — Summer 2025 - Spring 2026
Along with Robert Muth, advised research on the use of machine learning for understanding the relationship between Kleshchev multipartitions and MV-polytopes

Advisor for two students participating in DU's Undergraduate Research Program — Summer 2024
Projects involved designing and implementing an algorithm for finding a Seifert matrix for a knot given its PD notation and relating determinants of spiral knots to integer sequences.

Advisor of undergraduate research project on properties of twisted torus knots. — Spring 2024

Co-advisor of undergraduate research project for student Jacob Swenberg — Spring 2020
Along with Vladimir Chernov and Ina Petkova, advised research on link polynomials and causality in spacetimes. Resulted in a co-authored paper, listed above.

Mentor in the Directed Reading Program at Indiana University — Fall 2017

Outreach, Service, and Other Activities

Steering Committee Member for DU's Women in STEM organization — Ongoing
Representative of the School of Science and Engineering in DU's Faculty Senate Assembly — Ongoing
Secretary for the Allegheny Mountain section of the Mathematical Association of America — Ongoing
Committee member for the Department Awards Committee — Springs 2024-2026
Committee member for the School of Science and Engineering DEI Committee — Fall 2023 - Fall 2024
Panelist for a grant review panel at the National Science Foundation — Spring 2023
Organizer of Pi Day even hosted by the department and open to the university — March 2023, 2024
Organizer of departmental game nights for faculty and students — Springs 2023, 2024
Panelist at Pittsburgh Women in Mathematics and Computing Symposium (WMCS) — February 2023
Co-organizer of an MSRI Special Session at the Joint Mathematics Meeting — Winter 2023
Organizer of topic group at Nearly Carbon Neautral Geometric Topology Conference — September 2022
Co-organizer of the Dartmouth College Topology Seminar — Winter 2021
Co-host of the Math 8 Sessions of the Undergraduate Open House — September 2020
Presenter at The Governor's Institute of Vermont in Mathematical Sciences — June 2019, 2020
2019 topic: introduction to knot theory. 2020 topic: platonic solids, Euler, and induction.
Member of selection committee for the "Biographies of Contemporary Women in Mathematics" essay contest sponsored by the Dartmouth Mathematics Department — Winters 2019, 2020
Judge for the Dartmouth Math Department's Undergraduate Poster Session — Spring 2019
Co-organizer of Sonia Kovalesky Math Day at Dartmouth College — Spring 2019
Volunteer at IU Science Fest — Falls 2015, 2017
Volunteer at Math and Science Night at McCormick's Creek Elementary School — Spring 2017
Tutor in the Women in STEM Living-Learning Center — Spring 2014 – Spring 2015

Professional Development

Participant in online Philadelphia Area (Contact) Topology Seminar — Ongoing
Attended MAA Allegheny Mountain Section Meeting — Springs 2024-2026
Participant in Research Experience for Undergraduate Faculty at ICERM — June 2025
Participant in CTE's Inspired Teaching Retreat — May 2025
Participant in Braids Reunion Workshop at ICERM — July 2024
Attended conference on Combinatorial and Gauge theoretical methods in low dimensional topology and geometry at Centro di Ricerca de Matematica in Pisa, Italy — June 2024
Participant in CTE discussion on "Mentoring Underrepresented Students in STEMM" — March 2024
Attended CTE Micro-Workshop on Transparent Assignment Design — October 2023
Attended Allegheny Mountain Section NExT Workshop — September 2023
Participant in the CTE book study on Inclusive Teaching — Spring 2023
Attended CTE Micro-Workshop on Cultivating Student Feedback — February 2023
Certificate in Diversity and Inclusion at UGA (completed) — Spring 2022
Attended the Georgia Topology Conference — June 2021
Participant in the AIM 4-dimensional Topology virtual research community — Spring 2021
Attended the Simons Center workshop on Floer homology in low-dimensional topology — January 2021
Attended the Joint Mathematics Meetings of the AMS and MAA — January 2021
Attended the Tech Topology Conference (Georgia Tech) — December 2020
Attended the ICERM workshop Perspectives on Dehn Surgery — July 2019
Attended the Workshop in Geometric Topology (Calvin College) — June 2018
Attended the Joint Mathematics Meetings of the AMS and MAA — January 2018
Attended the UCLA Topology Conference 2018 — January 2018
Attended the Banff workshop on Thirty Years of Floer Theory for 3-Manifolds — July 2017

Programming and Other Skills

Proficient in LaTeX, Mathematica, Python
Familiar with MATLAB, C++
Teaching experience in the use of:
Canvas and Brightspace (LMS),
Zoom (for teaching & collaboration),
TechSmith Relay and Kaltura (for video capture and editing),
WeBWorK and WebAssign (online homework systems), and
Gradescope (for grading in the remote setting).